Math for Data Science – Exam – Fall 2025

NB: There are five sections in the exam. Indicative point scale, currently computed out of 40 points. No point without justification. ***Please give all documents at the end of the exam (including your draft paper and this document).*** *Good luck!*

1 Proofs – Vector spaces (5 pts)

1. **Sum of subspaces.** Let E be a vector space, F and G be two subspaces of E. We recall that their sum is defined by

$$F + G = \{u \in E, \exists (v, w) \in F \times G, u = v + w\}.$$

   (a) Is $F + G$ a subspace of E? Prove it.

   (b) Let suppose that $F \cap G = \{0_E\}$ (direct sum). Show that for every $u \in F + G$, there exists a unique pair $(v, w) \in F \times G$ such that $u = v + w$. (Hint: prove existence, then uniqueness).

2. **Injectivity and kernel.** Let E and F be two vector spaces and $f : E \to F$ be a linear map.

   (a) Prove that $f(0_E) = 0_F$.

   (b) Prove that f is injective *if and only if* $\mathrm{Ker}(f) = \{0_E\}$. (Hint: prove one implication, then the other).

Solution

1. **Sum of subspaces.** Let E be a vector space, F and G be two subspaces of E. Recall

$$F + G = \{u + v : \ u \in F, \ v \in G\}.$$

   (a) **(1pt)** *Yes, $F + G$ is a subspace of E.*

   Proof. First, $0_E = 0_E + 0_E \in F + G$, so $F + G \neq \varnothing$. Let $x_1 = v_1 + w_1$ and $x_2 = v_2 + w_2$ with $v_i \in F$, $w_i \in G$. Since F, G are subspaces, for any $\alpha, \beta \in \mathbb{K}$,

$$\alpha x_1 + \beta x_2 = \alpha(v_1 + w_1) + \beta(v_2 + w_2) = (\alpha v_1 + \beta v_2) + (\alpha w_1 + \beta w_2) \in F + G.$$

   Hence $F + G$ is closed under linear combinations; thus it is a subspace of E. □

   (b) *Uniqueness in the direct sum.* **(1pt)** Assume $F \cap G = \{0_E\}$. Let $u \in F + G$.

   Proof. By definition of $F + G$, there exist $v \in F$ and $w \in G$ with $u = v + w$ (existence). For uniqueness, suppose $u = v + w = v' + w'$ with $v, v' \in F$ and $w, w' \in G$. Then

$$v - v' = w' - w.$$

   The left-hand side lies in F, the right-hand side in G, hence $v - v' \in F \cap G = \{0_E\}$. Therefore $v = v'$ and then $w = w'$. This proves the uniqueness of the pair (v, w). □

2. **Injectivity and kernel.** Let E and F be vector spaces and $f : E \to F$ linear.

   (a) **(1pt)** $f(0_E) = 0_F$.

   Proof. Using additivity,

   $$f(0_E) = f(0_E + 0_E) = f(0_E) + f(0_E),$$

   hence $f(0_E) = 0_F$ after subtracting $f(0_E)$ from both sides. $\square$

   (b) **(2pt = 1pt for each implication)** f *is injective* $\iff \mathrm{Ker}(f) = \{0_E\}$.

   Proof. ($\Rightarrow$) If f is injective and $x \in \mathrm{Ker}(f)$, then $f(x) = 0_F = f(0_E)$, so by injectivity $x = 0_E$. Thus $\mathrm{Ker}(f) = \{0_E\}$.
   ($\Leftarrow$) Conversely, suppose $\mathrm{Ker}(f) = \{0_E\}$. If $f(x) = f(y)$, then $f(x - y) = 0_F$, hence $x - y \in \mathrm{Ker}(f) = \{0_E\}$, so $x = y$. Therefore f is injective. $\square$

2 Exercise – Determinant (6 pts)

Let $n \in \mathbb{N}^*$. We consider the determinant Δ_n defined by:

$$\Delta_n = \begin{vmatrix} 1 & 2 & 3 & \cdots & (n-1) & n \\ -1 & 0 & 3 & \cdots & (n-1) & n \\ -1 & -2 & 0 & \cdots & (n-1) & n \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ -1 & -2 & -3 & \cdots & 0 & n \\ -1 & -2 & -3 & \cdots & -(n-1) & 0 \end{vmatrix}$$

For example, we have:

$$\Delta_3 = \begin{vmatrix} 1 & 2 & 3 \\ -1 & 0 & 3 \\ -1 & -2 & 0 \end{vmatrix} \qquad \Delta_4 = \begin{vmatrix} 1 & 2 & 3 & 4 \\ -1 & 0 & 3 & 4 \\ -1 & -2 & 0 & 4 \\ -1 & -2 & -3 & 0 \end{vmatrix} \qquad \Delta_5 = \begin{vmatrix} 1 & 2 & 3 & 4 & 5 \\ -1 & 0 & 3 & 4 & 5 \\ -1 & -2 & 0 & 4 & 5 \\ -1 & -2 & -3 & 0 & 5 \\ -1 & -2 & -3 & -4 & 0 \end{vmatrix}$$

1. Compute Δ_1, Δ_2, Δ_3 and Δ_4. (Hint: use elementary operations on the rows and/or columns of the matrix.)
2. Make a hypothesis on the value of Δ_n for every $n \in \mathbb{N}^*$ (hint: use $n!$.). Explain your reasoning.
3. Prove your hypothesis by induction. (Hint: use your previous answers for initializing the proof. Then prove that if the property is true for a given n, then it is also true for $n+1$.)

Solution

1. Compute Δ_1 **(0.5pt)**, Δ_2 **(0.5pt)**, Δ_3**(1pt)**and Δ_4 **(1pt)**. **(i.e. total=3pt)** Either : compute the determinants directly. Or : For each determinant, perform the same determinant-preserving row operations: add the first row to every other row, i.e.

$$R_i \leftarrow R_i + R_1 \quad \text{for all } i = 2, \ldots, n.$$

This makes the first column equal to $(1, 0, \ldots, 0)^\top$, so we can expand along the first column and read off an upper triangular $(n-1) \times (n-1)$ block whose diagonal entries are $2, 3, \ldots, n$.

- $n = 1$: $\Delta_1 = \begin{vmatrix} 1 \end{vmatrix} = 1$.
- $n = 2$:
$$\Delta_2 = \begin{vmatrix} 1 & 2 \\ -1 & 0 \end{vmatrix} \xrightarrow{R_2 \leftarrow R_2 + R_1} \begin{vmatrix} 1 & 2 \\ 0 & 2 \end{vmatrix} = 1 \cdot 2 = 2.$$
- $n = 3$:
$$\Delta_3 = \begin{vmatrix} 1 & 2 & 3 \\ -1 & 0 & 3 \\ -1 & -2 & 0 \end{vmatrix} \xrightarrow{R_2, R_3 \leftarrow + R_1} \begin{vmatrix} 1 & 2 & 3 \\ 0 & 2 & 6 \\ 0 & 0 & 3 \end{vmatrix} = 1 \cdot 2 \cdot 3 = 6.$$

- $n = 4$:

$$\Delta_4 = \begin{vmatrix} 1 & 2 & 3 & 4 \\ -1 & 0 & 3 & 4 \\ -1 & -2 & 0 & 4 \\ -1 & -2 & -3 & 0 \end{vmatrix} \xrightarrow{R_2, R_3, R_4 \leftarrow + R_1} \begin{vmatrix} 1 & 2 & 3 & 4 \\ 0 & 2 & 6 & 8 \\ 0 & 0 & 3 & 8 \\ 0 & 0 & 0 & 4 \end{vmatrix} = 1 \cdot 2 \cdot 3 \cdot 4 = 24.$$

2. Make a hypothesis on the value of Δ_n for every $n \in \mathbb{N}^*$ (hint: use $n!$.). Explain your reasoning. **(1pt, only if explanation)** v1 (ok) : just remark that the computed values correspond to $n!$:

$$\Delta_1 = 1 = 1!, \quad \Delta_2 = 2 = 2!, \quad \Delta_3 = 6 = 3!, \quad \Delta_4 = 24 = 4!.$$

Hence we hypothesize that for every $n \in \mathbb{N}^*$,

$$\boxed{\Delta_n = n!}.$$

v2 (better) From the pattern above, after the row operations $R_i \leftarrow R_i + R_1$ ($i \geq 2$), the matrix becomes upper triangular once we expand along the first column, with diagonal entries $1, 2, 3, \ldots, n$. Hence

$$\boxed{\Delta_n = 1 \cdot 2 \cdot \cdots \cdot n = n!}.$$

3. **Proof by induction on n.** (Hint: use your previous answers for initializing the proof. Then prove that if the property is true for a given n, then it is also true for $n + 1$.) **(2pt = 0.5pt for initialization + 1.5pt for induction step). Give 2pt only if the proof is complete and correct. / Give 1pt if sketch of proof.**

initialization. From item (1), $\Delta_1 = 1 = 1!$, $\Delta_2 = 2 = 2!$, $\Delta_3 = 6 = 3!$, and $\Delta_4 = 24 = 4!$.

Inductive step. Assume $\Delta_n = n!$ for some $n \geq 1$. Consider Δ_{n+1}. Apply the determinant-preserving operations

$$R_i \leftarrow R_i + R_1 \quad \text{for } i = 2, \ldots, n + 1.$$

Then the first column becomes $(1, 0, \ldots, 0)^\top$, so expanding along that column shows

$$\Delta_{n+1} = \det \begin{pmatrix} 2 & 6 & 8 & \cdots & 2(n+1) \\ 0 & 3 & 8 & \cdots & 2(n+1) \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & n & 2(n+1) \\ 0 & \cdots & \cdots & 0 & n+1 \end{pmatrix} = 2 \cdot 3 \cdot \cdots \cdot (n+1).$$

Hence

$$\Delta_{n+1} = (n+1)(2 \cdot 3 \cdot \cdots \cdot n) = (n+1)\Delta_n.$$

By the induction hypothesis $\Delta_n = n!$, we obtain

$$\Delta_{n+1} = (n+1)\, n! = (n+1)!.$$

Thus, by induction, $\Delta_n = n!$ for all $n \in \mathbb{N}^*$. $\square$

3 Proofs – Eigenvalues and eigenvectors (6 pts)

Let $A \in \mathbb{R}^{n\times n}$ and suppose (λ_i, v_i), $i = 1, \ldots, n$, are eigenpairs of A with the eigenvectors $v_1, \ldots, v_n$ linearly independent. We set $V = [v_1 \ \cdots \ v_n] \in \mathbb{R}^{n\times n}$ (columns of V are the v_i vectors) and $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n) \in \mathbb{R}^{n\times n}$.

1. (a) Prove by induction that for every integer $k \geq 1$, we have:

$$\forall i \in \mathbb{N}, 1 \leq i \leq n, A^k v_i = \lambda_i^k v_i,$$

   (b) Deduce the matrix identity:

$$A^k = VD^kV^{-1}.$$

2. Now assume A is invertible.

   (a) What does it mean for the eigenvalues of A? and for D?

   (b) Prove that for each $i, 1 \leq i \leq n$, v_i is an eigenvector of A^{-1} associated with the eigenvalue λ_i^{-1}, i.e.,

$$A^{-1}v_i = \lambda_i^{-1}v_i,$$

   (c) Deduce the matrix identity:

$$A^{-1} = VD^{-1}V^{-1}.$$

3. Are A^k (for $k \geq 1$) and A^{-1} (when defined) diagonalizable? Justify based on the previous questions.

Solution

1. (a) *Proof by induction on k.* **(1pt)**

   Proof. For $k = 1$, the result holds by definition of eigenpair. Assume it holds for some $k \geq 1$. Then

$$A^{k+1}v_i = A(A^k v_i) = A(\lambda_i^k v_i) = \lambda_i^k(Av_i) = \lambda_i^k(\lambda_i v_i) = \lambda_i^{k+1}v_i.$$

   This completes the induction. □

   (b) *Matrix identity.* **(1pt)**

   Proof. Since the v_i are linearly independent, V is invertible. For each i, we have

$$A^k v_i = \lambda_i^k v_i = VD^k e_i,$$

   where e_i is the i-th standard basis vector. Thus $A^kV = VD^k$, and multiplying on the right by V^{-1} gives the result. □

2. Now assume A is invertible.

   (a) *Eigenvalues and diagonal entries.* **(0.5 + 0.5 pt)** If A is invertible, then none of its eigenvalues λ_i can be zero, since otherwise there would exist a nonzero vector v_i with $Av_i = 0$. Thus all diagonal entries of D are nonzero, and hence D is invertible.

(b) *Eigenvectors of A^{-1}.* **(1pt)**

Proof. For each i, since $\lambda_i \neq 0$, we have

$$A^{-1}v_i = A^{-1}(Av_i/\lambda_i) = (1/\lambda_i)(A^{-1}A)v_i = (1/\lambda_i)v_i.$$

Thus, v_i is an eigenvector of A^{-1} with eigenvalue λ_i^{-1}. □

(c) *Matrix identity for A^{-1}.* **(1pt)**

Proof. Using the result from part (a), we have

$$A^{-1}V = VD^{-1},$$

and multiplying on the right by V^{-1} gives the desired identity. □

3. *Diagonalizability of A^k and A^{-1}.* **(0.5 + 0.5pt / 0.5 if only one is justified)** Yes, both A^k and A^{-1} are diagonalizable. This follows from the identities

$$A^k = VD^kV^{-1} \quad \text{and} \quad A^{-1} = VD^{-1}V^{-1},$$

where D^k and D^{-1} are diagonal matrices. Since V is invertible, both A^k and A^{-1} are diagonalizable.

4 Computations – Diagonalization (9 pts)

We consider the matrix

$$A = \begin{pmatrix} 3 & -1 & 1 \\ 0 & 2 & 2 \\ -1 & 1 & 3 \end{pmatrix}$$

We want to know whether it is diagonalizable in $\mathcal{M}_3(\mathbb{R})$. We recall that the algebraic multiplicity of an eigenvalue λ is its multiplicity as a root of the characteristic polynomial of A, and the geometric multiplicity of λ is the dimension of the eigenspace associated to λ:

$$\mathrm{Eig}(A, \lambda) = \{X \in \mathbb{R}^3 : \ AX = \lambda X\}.$$

1. (a) *Using the characteristic polynomial of* A, determine the eigenvalues of A and their algebraic multiplicities. (Hint: the characteristic polynomial that you obtain should be divisible by $P(\lambda) = (\lambda - 2)(\lambda - 4)$.)
   (b) Using a property of the trace, check your previous answer.
2. For each eigenvalue, determine *a basis* of the corresponding eigenspace, and the geometric multiplicity of this eigenvalue.
3. Based on your previous answers, is A diagonalizable in $\mathcal{M}_3(\mathbb{R})$? Justify your answer.

Solution

1. (a) **Eigenvalues and algebraic multiplicities. (0.5pt for writing the characteristic polynomial as a determinant, 0.5pt for computing the determinant, 0.5+0.5pt for giving the eigenvalues = roots of the characteristic polynomial, 0.5+0.5pt for algebraic multiplicities; total = 3pt)** To determine the eigenvalues of A, we form the characteristic polynomial:

$$\chi_A(\lambda) = \det \begin{pmatrix} 3-\lambda & -1 & 1 \\ 0 & 2-\lambda & 2 \\ -1 & 1 & 3-\lambda \end{pmatrix} \overset{C_1 \leftarrow C_1 + C_2}{=} \det \begin{pmatrix} 2-\lambda & -1 & 1 \\ 2-\lambda & 2-\lambda & 2 \\ 0 & 1 & 3-\lambda \end{pmatrix}$$

$$\overset{L_2 \leftarrow L_2 - L_1}{=} \det \begin{pmatrix} 2-\lambda & -1 & 1 \\ 0 & 3-\lambda & 1 \\ 0 & 1 & 3-\lambda \end{pmatrix} = (2-\lambda) \det \begin{pmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{pmatrix}$$

$$= (2-\lambda)((3-\lambda)^2 - 1) = -(\lambda-2)^2(\lambda-4).$$

Hence the eigenvalues of A are 2 (algebraic multiplicity 2) and 4 (algebraic multiplicity 1).

Other way of computing $\chi_A(\lambda)$: Expanding along the first row:

$$\chi_A(\lambda) = (3-\lambda)((2-\lambda)(3-\lambda) - 2) + (-1)\cdot(-1)\cdot \underbrace{\begin{vmatrix} 0 & 2 \\ -1 & 3-\lambda \end{vmatrix}}_{=2} + 1 \cdot \underbrace{\begin{vmatrix} 0 & 2-\lambda \\ -1 & 1 \end{vmatrix}}_{=2-\lambda}.$$

Hence

$$\chi_A(\lambda) = (3-\lambda)\left(\lambda^2 - 5\lambda + 4\right) + 2 + (2-\lambda) = -\lambda^3 + 8\lambda^2 - 20\lambda + 16 = -(\lambda-4)(\lambda-2)^2.$$

(b) **Check with the trace. (0.5pt for reminding the property, 0.5pt for correct check; total = 1pt)** As a check, since χ_A splits over $\mathbb{R}$, the sum of the eigenvalues (with multiplicities) is $2+2+4=8$, which equals $\operatorname{tr}(A)=3+2+3=8$.

2. **Eigenspaces and geometric multiplicities. (2pt for each eigenvalue; total = 4pt; for each eigenvalue, split as 0.5pt for writing $A-\lambda I$ / starting the solving, 0.5pt for solving the system, 0.5pt for basis vector, 0.5pt for geometric multiplicity)**

   - For $\lambda = 4$:
   $$A - 4I = \begin{pmatrix} -1 & -1 & 1 \\ 0 & -2 & 2 \\ -1 & 1 & -1 \end{pmatrix}.$$
   Solving $(A-4I)x = 0$ gives $x = 0$, $y = z$; a basis vector is $(0,1,1)^\top$. Thus
   $$\operatorname{Eig}(A,4) = \operatorname{span}\{(0,1,1)\}, \qquad \dim \operatorname{Eig}(A,4) = 1.$$
   - For $\lambda = 2$:
   $$A - 2I = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 0 & 2 \\ -1 & 1 & 1 \end{pmatrix}.$$
   Solving $(A-2I)x = 0$ yields $z = 0$ and $y = x$; a basis vector is $(1,1,0)^\top$. Hence
   $$\operatorname{Eig}(A,2) = \operatorname{span}\{(1,1,0)\}, \qquad \dim \operatorname{Eig}(A,2) = 1.$$

Other writing style : For $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3$,

$$\begin{aligned} X \in \operatorname{Eig}(A,2) &\iff AX = 2X \\ &\iff \begin{cases} 3x - y + z = 2x, \\ 2y + 2z = 2y, \\ -x + y + 3z = 2z \end{cases} \iff \begin{cases} x - y + z = 0, \\ z = 0, \\ -x + y + z = 0. \end{cases} \end{aligned}$$

Thus $x = y$ and $z = 0$, so

$$\operatorname{Eig}(A,2) = \operatorname{Span}\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\}, \qquad \dim \operatorname{Eig}(A,2) = 1.$$

3. **Diagonalizability. (1pt only if justified correctly)** Since for the eigenvalue 2 the geometric multiplicity is $1 < 2$ (its algebraic multiplicity), A is *not* diagonalizable over $\mathbb{R}$.

   Equivalently : The sum of the geometric multiplicities is
   $$\dim \operatorname{Eig}(A,4) + \dim \operatorname{Eig}(A,2) = 1 + 1 = 2 < 3.$$
   Therefore A is *not* diagonalizable in $\mathcal{M}_3(\mathbb{R})$.

5 Exercise – Orthogonality between $S_n(\mathbb{R})$ and $A_n(\mathbb{R})$ (14 pts)

We denote by $\mathcal{M}_n(\mathbb{R})$ the set of square matrices of size n with real entries, by $S_n(\mathbb{R})$ the subset of symmetric matrices and by $A_n(\mathbb{R})$ the subset of antisymmetric matrices.
Let $n \in \mathbb{N}^*$. We consider the application from $\mathcal{M}_n(\mathbb{R}) \times \mathcal{M}_n(\mathbb{R})$ to $\mathbb{R}$ defined by:

$$(M, N) \longmapsto (M \mid N) = \operatorname{tr}\left(M^\top N\right).$$

1. **Inner product and norm.**
   (a) Show that $(\cdot \mid \cdot)$ is an inner product on $\mathcal{M}_n(\mathbb{R})$.
   (b) Deduce that the function from $\mathcal{M}_n(\mathbb{R})$ to $\mathbb{R}$ defined by $M \mapsto \|M\| = \sqrt{\operatorname{tr}(M^\top M)}$ is a norm on $\mathcal{M}_n(\mathbb{R})$.
   (c) We consider the standard basis of $\mathcal{M}_n(\mathbb{R})$ defined by the matrices E_{ij}, $i, j \in [\![1, n]\!]$, where E_{ij} is the matrix with a 1 at entry (i, j) and 0 elsewhere. Show that this basis is orthonormal for the above defined inner product.

2. **Application.**
   (a) Let
   $$W = \{A \in \mathcal{M}_3(\mathbb{R}) : \ \operatorname{tr}(A) = 0\}.$$
   We remark that W is a subspace of $\mathcal{M}_3(\mathbb{R})$. Using the above defined inner product on $\mathcal{M}_3(\mathbb{R})$, compute $W^\perp$, the orthogonal complement of W.
   (b) What are the dimensions of W and $W^\perp$? Check your answer with a property from the course.

3. **Subspaces.**
   (a) Define $S_n(\mathbb{R})$ and $A_n(\mathbb{R})$ using a property of the transpose (i.e. with a set notation).
   (b) Show that $S_n(\mathbb{R})$ and $A_n(\mathbb{R})$ are two subspaces in $\mathcal{M}_n(\mathbb{R})$.
   (c) Show that $S_n(\mathbb{R})$ and $A_n(\mathbb{R})$ are complementary and orthogonal subspaces in $\mathcal{M}_n(\mathbb{R})$.
   (d) What is the dimension of $S_n(\mathbb{R})$ and $A_n(\mathbb{R})$? Propose a basis for each of these subspaces.

4. **Distance to $S_n(\mathbb{R})$ (changed to full bonus question).** We recall that for a subspace F of a normed vector space $(E, \|\cdot\|)$, we define the distance from a point $x \in E$ to F by
   $$d(x, F) = \inf_{y \in F} \|x - y\|.$$
   (a) Show that for every $M \in \mathcal{M}_n(\mathbb{R})$, there exists a unique matrix $S \in S_n(\mathbb{R})$ such that $d(M, S_n(\mathbb{R})) = \|M - S\|$.
   (b) For every $M \in \mathcal{M}_n(\mathbb{R})$, compute the distance $d(M, S_n(\mathbb{R}))$ as a function of M.
   (c) For $i, j \in [\![1, n]\!]$, we denote by E_{ij} the matrix of $\mathcal{M}_n(\mathbb{R})$ with a 1 at entry (i, j) and 0 elsewhere. We consider $M = \sum_{i=1}^{n} E_{i1}$. Compute $d(M, S_n(\mathbb{R}))$.

Solution

Let $M_n(\mathbb{R})$ be the space of $n \times n$ real matrices and define

$$(M, N) \ \longmapsto \ (M \mid N) \coloneqq \operatorname{tr}(M^\top N).$$

1. **Inner product and norm. (total = 3pt)**

   (a) $(\cdot \mid \cdot)$ **is an inner product on** $M_n(\mathbb{R})$**. (2pt if all properties are proved correctly; 1pt if only 2 properties are proved correctly, 0.5pt if only 1 property is proved correctly)**
   Bilinearity. For $\alpha, \beta \in \mathbb{R}$ and $M_1, M_2, N \in M_n(\mathbb{R})$,

   $$(\alpha M_1 + \beta M_2 \mid N) = \operatorname{tr}\big((\alpha M_1 + \beta M_2)^\top N\big) = \alpha \operatorname{tr}(M_1^\top N) + \beta \operatorname{tr}(M_2^\top N) = \alpha(M_1 \mid N) + \beta(M_2 \mid N).$$

   Linearity in the second slot is analogous.
   Symmetry. Using $\operatorname{tr}(X) = \operatorname{tr}(X^\top)$,

   $$(M \mid N) = \operatorname{tr}(M^\top N) = \operatorname{tr}\big((M^\top N)^\top\big) = \operatorname{tr}(N^\top M) = (N \mid M).$$

   Positive definiteness. We have

   $$(M \mid M) = \operatorname{tr}(M^\top M) = \sum_{i,j=1}^{n} M_{ij}^2 \geq 0,$$

   and $(M \mid M) = 0$ implies $M_{ij} = 0$ for all i, j, hence $M = 0$.
   Therefore $(\cdot \mid \cdot)$ is an inner product (the Frobenius inner product).

   (b) $\|M\| := \sqrt{(M \mid M)}$ **is a norm. (1pt if justification based on the inner product, or 1pt properties of the norm are checked correctly)**
   Since it is induced by an inner product, it satisfies:

   $$\|M\| \geq 0, \quad \|M\| = 0 \iff M = 0, \quad \|\alpha M\| = |\alpha|\,\|M\|,$$

   and by the polarization/Cauchy–Schwarz inequality,

   $$\|M + N\|^2 = \|M\|^2 + 2(M \mid N) + \|N\|^2 \leq \big(\|M\| + \|N\|\big)^2,$$

   hence the triangle inequality. Thus $\|\cdot\|$ is a norm (the Frobenius norm).

   (c) **The standard basis** $\{E_{ij}\}$ **is orthonormal. (0.5pt orthogonality + 0.5pt normalization; total = 1pt)**

2. **Application. (total = 3pt)**

   (a) **Computation of** $W^\perp$**. (2pt, give 1pt for starting, 1pt for correct final answer)**
   First, $\langle A, I_3 \rangle = \operatorname{tr}(A^\top I_3) = \operatorname{tr}(A)$, so $I_3 \perp W$ and hence $\operatorname{span}\{I_3\} \subseteq W^\perp$.
   Conversely, take $X \in W^\perp$. Decompose $X = \alpha I_3 + Y$ with $\alpha = \frac{1}{3}\operatorname{tr}(X)$ so that $\operatorname{tr}(Y) = 0$, i.e. $Y \in W$. Then

   $$0 = \langle X, Y \rangle = \langle \alpha I_3 + Y,\ Y \rangle = \alpha \underbrace{\langle I_3, Y \rangle}_{\operatorname{tr}(Y)=0} + \langle Y, Y \rangle = \|Y\|_F^2,$$

   hence $Y = 0$ and $X = \alpha I_3$. Therefore $W^\perp = \operatorname{span}\{I_3\}$.
   Other writing style possible (more detailed / bof)
   For $M \in M_3(\mathbb{R})$, we have

   $$M \in W^\perp \iff \forall A \in W,\ (M \mid A) = 0 \iff \forall A \in W,\ \operatorname{tr}(M^\top A) = 0.$$

Since $\mathrm{tr}(M^\top A) = \mathrm{tr}(AM^\top)$, and $\mathrm{tr}(XY) = \mathrm{tr}(YX)$ for any $X, Y \in M_3(\mathbb{R})$, we have

$$M \in W^\perp \iff \forall A \in W,\ \mathrm{tr}(AM^\top) = 0.$$

Now, for $A \in W$, we have $\mathrm{tr}(A) = 0$, so

$$\mathrm{tr}(AM^\top) = \mathrm{tr}(A(M^\top - \tfrac{\mathrm{tr}(M)}{3} I_3)) + \tfrac{\mathrm{tr}(M)}{3}\mathrm{tr}(A) = \mathrm{tr}(A(M^\top - \tfrac{\mathrm{tr}(M)}{3} I_3)).$$

Thus,

$$M \in W^\perp \iff \forall A \in W,\ \mathrm{tr}\Big(A\left(M^\top - \tfrac{\mathrm{tr}(M)}{3} I_3\right)\Big) = 0.$$

Since A is arbitrary in W, this holds if and only if

$$M^\top - \tfrac{\mathrm{tr}(M)}{3} I_3 = 0,$$

i.e., $M = \frac{\mathrm{tr}(M)}{3} I_3$ is a scalar multiple of the identity matrix. Therefore,

$$W^\perp = \{\alpha I_3 :\ \alpha \in \mathbb{R}\}.$$

$$\boxed{W^\perp = \mathrm{span}\{I_3\}} \qquad (\text{and } \dim W = 8,\ \dim W^\perp = 1).$$

(b) **Dimension check (1pt)** We have $\dim(W^\perp) = 1$ since it is spanned by I_3. Since $\dim(M_3(\mathbb{R})) = 9$, we have, by the dimension theorem for orthogonal complements,

$$\dim(W) = \dim(M_3(\mathbb{R})) - \dim(W^\perp) = 9 - 1 = 8.$$

This is consistent with the fact that W is defined by a single linear equation $\mathrm{tr}(A) = 0$. NB: other writing style possible using the rank-nullity theorem.

3. **Subspaces. (total = 7pt)**

(a) **Definition of $S_n(\mathbb{R})$ (0.5pt) and $A_n(\mathbb{R})$ (0.5pt), total=1pt.** We have

$$S_n(\mathbb{R}) = \{M \in M_n(\mathbb{R}) :\ M^\top = M\}, \qquad A_n(\mathbb{R}) = \{M \in M_n(\mathbb{R}) :\ M^\top = -M\}.$$

(b) **$S_n(\mathbb{R})$ and $A_n(\mathbb{R})$ are subspaces. (1pt + 1pt = total 2pt)** The zero matrix is symmetric and antisymmetric. If S_1, S_2 are symmetric then $(\alpha S_1 + \beta S_2)^\top = \alpha S_1 + \beta S_2$ is symmetric, so $S_n(\mathbb{R})$ is stable by linear combination, hence it is a subspace. Similarly, if A_1, A_2 are antisymmetric ($A_i^\top = -A_i$), then $(\alpha A_1 + \beta A_2)^\top = -\alpha A_1 - \beta A_2$ which is antisymmetric, so $A_n(\mathbb{R})$ is a subspace.

(c) **$S_n(\mathbb{R})$ and $A_n(\mathbb{R})$ are orthogonal complements. (1pt for decomposition + 1pt for orthogonality = total 2pt)**
For any $M \in M_n(\mathbb{R})$, let us consider

$$S = \frac{M + M^\top}{2} \in S_n(\mathbb{R}), \qquad A = \frac{M - M^\top}{2} \in A_n(\mathbb{R}).$$

Then $M = S + A$.
If $X \in S_n(\mathbb{R}) \cap A_n(\mathbb{R})$, then $X^\top = X = -X$, hence $X = 0$, so the decomposition is unique and $M_n(\mathbb{R}) = S_n(\mathbb{R}) \oplus A_n(\mathbb{R})$.
Orthogonality follows from

$$(S \mid A) = \mathrm{tr}(S^\top A) = \mathrm{tr}(SA) = \mathrm{tr}(A^\top S) = \mathrm{tr}((-A)S) = -\mathrm{tr}(AS) = -(S \mid A),$$

hence $(S \mid A) = 0$. Thus $S_n(\mathbb{R}) \perp A_n(\mathbb{R})$ and they are complementary orthogonal subspaces.

(d) **Dimensions and bases of $S_n(\mathbb{R})$ and $A_n(\mathbb{R})$. (1pt for each subspace, split as 0.5pt each for intuition, 1pt each for clean proof with basis vectors, total = 2pt)** Let $E_{ij} \in M_n(\mathbb{R})$ denote the matrix with a 1 in position (i, j) and 0 elsewhere. We can remember that $\{E_{ij}\}_{1\leq i,j\leq n}$ is a basis of $M_n(\mathbb{R})$ with n^2 elements. But let us check it by proving the following claim: *The family $\{E_{ij}\}_{1\leq i,j\leq n}$ is orthonormal for $(\cdot \mid \cdot)$.* Indeed, for $1 \leq i, j, k, l \leq n$,

$$(E_{ij} \mid E_{kl}) = \mathrm{tr}(E_{ij}^\top E_{kl}) = \mathrm{tr}(E_{ji}E_{kl}) = \mathrm{tr}(\delta_{jk}E_{il}) = \delta_{jk}\mathrm{tr}(E_{il}) = \delta_{jk}\delta_{il}.$$

Since $\{E_{ij}\}_{i,j}$ is a basis of $M_n(\mathbb{R})$ with n^2 elements, it is an orthonormal basis.

Now we want to build bases of $S_n(\mathbb{R})$ and $A_n(\mathbb{R})$. We can use the decomposition $M = S + A$ above to find the dimensions of these subspaces. In particular, we can consider the matrices of the form $E_{ij} + E_{ji}$ for $1 \leq i < j \leq n$ and the matrices E_{ii} for $1 \leq i \leq n$ to build a basis of $S_n(\mathbb{R})$. Similarly, we can consider the matrices of the form $E_{ij} - E_{ji}$ for $1 \leq i < j \leq n$ to build a basis of $A_n(\mathbb{R})$.

We can also directly remark that a symmetric matrix is determined by the n diagonal entries and the $\frac{n(n-1)}{2}$ entries above the diagonal, hence $\dim(S_n(\mathbb{R})) = n + \frac{n(n-1)}{2} = \frac{n(n+1)}{2}$. A basis is given by the matrices E_{ii} for $1 \leq i \leq n$ (with a 1 at (i, i) and 0 elsewhere) and the matrices $E_{ij} + E_{ji}$ for $1 \leq i < j \leq n$.

Similarly, an antisymmetric matrix is determined by the $\frac{n(n-1)}{2}$ entries above the diagonal (since the diagonal entries are 0 (!!) and the entries below the diagonal are the opposite of those above), hence $\dim(A_n(\mathbb{R})) = \frac{n(n-1)}{2}$. A basis is given by the matrices $E_{ij} - E_{ji}$ for $1 \leq i < j \leq n$.

Finally, we check that

$$\dim(S_n(\mathbb{R})) + \dim(A_n(\mathbb{R})) = \frac{n(n+1)}{2} + \frac{n(n-1)}{2} = n^2 = \dim(M_n(\mathbb{R})).$$

4. **Distance to $S_n(\mathbb{R})$ (BONUS, 4pt = 1 + 1 + 2).**

(a) Justify the existence and uniqueness of the projection. **(1pt)** Since $S_n(\mathbb{R})$ is a subspace of the inner product space $(M_n(\mathbb{R}), (\cdot \mid \cdot))$, for every $M \in M_n(\mathbb{R})$, there exists a unique $S \in S_n(\mathbb{R})$ such that

$$\|M - S\| = \inf_{T\in S_n(\mathbb{R})} \|M - T\| = d(M, S_n(\mathbb{R})).$$

(b) **Computation of $d(M, S_n(\mathbb{R}))$. (1pt)** For $M \in M_n(\mathbb{R})$, we write $M = S + A$ as defined above. Let T be a symmetric matrix, we have:

$$\|M - T\|^2 = \|A + (S - T)\|^2 = \|A\|^2 + \|S - T\|^2$$

because $A \perp (S - T)$. Hence the minimum is attained at $T = S = \frac{M+M^\top}{2}$ and

$$d(M, S_n(\mathbb{R})) = \left\| \frac{M - M^\top}{2} \right\|.$$

(c) *Application (2pt).* Let E_{ij} denote the matrix with a 1 at (i, j) and 0 elsewhere, and take

$$M = \sum_{i=1}^{n} E_{i1}.$$

Then $M^\top = \sum_{i=1}^n E_{1i}$ and

$$A = \frac{M - M^\top}{2} = \frac{1}{2}\sum_{i=1}^{n}(E_{i1} - E_{1i}) = \frac{1}{2}\sum_{i=2}^{n}(E_{i1} - E_{1i}),$$

since the $i = 1$ term vanishes. The family $\{E_{ij}\}_{i,j}$ is orthonormal for $(\cdot \mid \cdot)$, hence

$$\|A\|^2 = \frac{1}{4}\sum_{i=2}^{n}\left(\|E_{i1}\|^2 + \|E_{1i}\|^2\right) = \frac{1}{4}\cdot 2(n-1) = \frac{n-1}{2}.$$

Therefore

$$d\big(M, S_n(\mathbb{R})\big) = \|A\| = \sqrt{\frac{n-1}{2}}.$$